Ajay Kumar Dwivedi

Hyderabad, Telangana, India | (+91) 986-853-4271 | ajay.dwivedi2007@gmail.com
ajaydwivedi.com | github.com/imajaydwivedi | linkedin.com/in/imajaydwivedi

Profile

Site Reliability Engineer with 15+ years of experience automating infrastructure through DevOps CI/CD pipelines, using scripting tools such as PowerShell, Python, Ansible, GitHub Actions, and containers. I build monitoring and alerting systems with Python, PowerShell, Flask, Prometheus, and Grafana for both on-premises and cloud platforms (AWS & Azure).

Core Competencies

- **Leadership & Reliability:** Leadership & Stakeholder Management, Reliability Engineering, Problem Management, On-Call, Capacity Planning, Customer-Facing
- **Databases:** SQL Server, PostgreSQL, MongoDB, Performance Tuning, Query Optimization, High Availability & DR
- **Cloud & Containers:** AWS, Azure, Cloud Infrastructure, EC2, RDS, Kubernetes, Distributed Systems
- **Automation & DevOps:** Infrastructure as Code, Automation, Terraform, Ansible, CI/CD, Deployment Pipelines, GitHub Actions, Python, Shell Scripting, Open-Source Development
- **Monitoring & AI:** Monitoring & Observability, Prometheus, Grafana, AI Agents (LangChain, Ollama)

Skills

- **Tools and Frameworks:** Git, Linux, Ansible, Flask, GitHub Actions, VS Code, Grafana, Prometheus, Containers, Docker, Ubuntu, RHEL, PowerShell DSC, Infrastructure as Code (IaC)
- **Programming Languages:** SQL, PowerShell, Python, JavaScript, HTML / CSS, Shell Scripting
- **Cloud Platforms:** AWS, Azure
- **High Availability & Disaster Recovery:** Clustering, Availability Groups, Mirroring, Log Shipping, Replication, Disk Stripping, RAID, Multi-Subnet Setup
- **Databases:** SQL Server, PostgreSQL, SQLite, RDS, Azure SQL, Managed Instance, Aurora
- **Languages:** English (proficient), Hindi (native)

Experience

Tessell *February 2026 – September 2026*
Principal Solution Architect - SQL Server

- Worked on Tessell's Database as a Service (DBaaS) platform, a distributed systems architecture with Docker/Kubernetes as the control plane and AWS, Azure, and GCP as the data plane
- Wrote Python-based SAGA workflows on Kubernetes and a dockerized environment to build Tessell's DBaaS platform on AWS and Azure, reducing provisioning time to 20 minutes
- Used root cause analysis to debug production issues and implement customer fixes during 24x7 on-call support
- Designed CentOS-to-RHEL migration automation for PostgreSQL using Python and Ansible, migrating 180+ database clusters with 15 minutes of downtime each
- Built DBaaS capabilities for deploying SQL Server on AWS EC2 and Azure VMs, enabling self-service concurrent deployments
- Added monitoring for data and log file growth using SQLExporter, Prometheus, and VictoriaMetrics
- Added enhancements to support Change Data Capture (CDC) and transactional replication

AngelOne *February 2022 – February 2026*
Senior Site Reliability Engineer (DevOps)

- Automated common database team activities, such as installation, configuration, monitoring, and alerting, using PowerShell, Flask, Python, T-SQL, Grafana, and PagerDuty for on-premises and cloud Managed Instances
- Improved database reliability and business continuity by automating ScriptOut of entire SQL Server instances for disaster recovery
- Automated AWS RDS/Aurora PostgreSQL database deployment and maintenance using Terraform
- Created a GitHub repo-based approach to baselining servers, simplifying monitoring with Prometheus and Grafana
- Built the SQLMonitor tool with dashboards that supported capacity planning and estimated storage needs for the next couple of years based on past growth
- Automated standalone PostgreSQL setup using Ansible
- Automated disk, connectivity, Patroni health, and backup alerts for PostgreSQL servers using Python
- Built Deployment Pipelines in GitHub Actions, using Ansible in the background, to set up multi-datacenter HA with a 3-node Patroni PostgreSQL cluster in the primary data center and a 3-node Patroni cluster in the secondary data center, linked by cascading streaming replication managed by Patroni, saving 80 hours/week of SRE effort with a single-click workflow
- Built a DB AI agent with LangChain and Ollama (local LLMs) that answers natural-language SQL Server performance questions by querying SQLMonitor data
- Baselined all Windows and Linux servers using Prometheus and Grafana
- Ran Problem Management for P1/P2 database incidents in Jira, performing an RCA for each database incident and representing the DBA team in CAB meetings

Arcesium *May 2020 – February 2022*
Database Reliability Engineer

- Applied reliability engineering as a Database Reliability Engineer, upgrading 30 critical SQL clusters/AGs in 12 hours to avoid a week-long downtime
- Resolved hundreds of incidents by moving 38 TB of AG databases with PowerShell
- Improved database reliability by automating ScriptOut of entire SQL Server instances for disaster recovery
- Built automation to prevent common issues such as tempdbSaver and LogSaver

TiVo *February 2018 – May 2020*
Database Administrator

- Implemented monitoring and alerting with Datadog
- Reduced performance incidents by 80% by building Delta Index Defragmentation automation for replicated and HADR databases
- Saved 40+ hours by automating SQL Server deployment and configuration using PowerShell
- Reduced transactional replication issues by 80% using latency monitoring infrastructure

United Health Group *October 2016 – February 2018*
Senior SQL DBA

- Handled incident management for production issues during 24x7 on-call support, practicing stakeholder management by directly interacting with customers to debug and fix critical customer-facing performance and deployment issues
- Saved $1 million per year through DB space capacity automation
- Improved performance issues by 50% and reduced VLDB maintenance time

Ericsson Global *June 2015 – October 2016*
Technical Lead

- Achieved a 3x performance improvement through advanced query optimization, such as recursive CTEs and table partitioning
- Led 18+ developers in following best practices and guidelines

TATA Consultancy Services *March 2011 – June 2015*
SQL DBA Developer

- Saved 20+ hours/week by automating core DBA tasks using dynamic SQL
- Moved 250+ large tables, each with billions of records, using SSIS in 45 hours
- Built data warehouse ETL pipelines using Pentaho and Kettle for high-volume liquor retail stores

Projects

SQLMonitor *January 2021 – Present*
Open-source, enterprise-grade SQL Server monitoring tool (GitHub)

- Built using T-SQL, PowerShell, Python, and Grafana
- Monitoring 500+ SQL Servers across environments
- Sends observability alerts to Slack, PagerDuty, and email
- Built-in auto-actions to clear and resolve issues

SQLDBATools *July 2021 – February 2023*
PowerShell module with cmdlets for SQL Server DBA activities (GitHub)

- Covers server inventory, disk and space estimation, backup history, permission audits, and maintenance job setup, reducing manual DBA effort by 60
- Built a monitoring and alerting system with Grafana dashboards, auto-clearing alerts, email notifications, and alert suppression
- Published to PSGallery for installation with `Install-Module`

AI Agent *October 2025 – Present*
LLM-powered SQL agent that answers natural-language performance questions about SQL Server instances monitored by SQLMonitor (GitHub)

- Built using Python, LangChain, Ollama, Streamlit, and SQLMonitor
- Grounds every answer in SQL queries against SQLMonitor data, shows the query in each explanation, and includes prompt guardrails against write statements, enabling application teams to troubleshoot performance problems themselves

HadrSync *January 2021 – Present*
PowerShell module that synchronizes all objects across replicas in Availability Groups (GitHub)

- Ensures zero sync issues during failovers and DR events
- Detects and synchronizes delta changes, such as changes to passwords, definitions, and more

Certifications

- Microsoft Certified Professional - Database Querying (October 2013)
- Microsoft Certified Professional - Database Administration (December 2013)
- BrentOzarUnlimited Mastering SQL Server (November 2022)
- SQLSkills Blackbelt Professional (June 2023)

Education

Bachelor of Engineering in Electronics & Communication Engineering *July 2006 – June 2010*
RustamJi Institute of Technology, Gwalior, Madhya Pradesh

Awards and Honors

- **Tech Titan**, AngelOne for 10x performance optimization on production SQL Server (2022)
- **Sapphire**, United Health Group for innovation and automation (2017)

Availability

Notice Period: 7 Days | Open to Relocation | Open to remote opportunities